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Remote Sensing Change Detection via a Spatial–Temporal Attention-Driven CNN–Transformer Architecture

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***Abstract*—In this study, we examine the efficacy of a model in identifying variations in satellite imagery before and after event occurrences. Our evaluation of the model is based on its ability to detect and predict change areas by comparing pre-event and post-event imagery along with other performance measures, including precision, recall, F1-score, accuracy, and Intersection over Union (IoU). Confusion matrices and evaluation outcomes are used to determine the weaknesses and strengths of the model, especially with regard to false positives and false negatives. The findings demonstrate a high level of accuracy but also reveal room for improvement in recall and precision to achieve better results in identifying minor changes. The results provide insights into future improvements of change detection models in remote sensing applications.**

***Index Terms*—remote sensing change detection, Land use change, Change detection, CBAM**

I. Introduction

The challenge of change detection is vital to geospatial science and ecological investigation , as it finds application in studying geographical formations and alterations occurring on the Earth's surface, which can be accomplished through satellite imaging or aerial photography . This technology has been widely adopted across various industries, including disaster management [1], urban planning , environmental monitoring [2], agricultural research , and land-use change detection .

The fundamental premise of change detection is to identify and measure variations between two or more collections of images captured at distinct times. These alterations could result from natural processes such as deforestation, flooding, or wildfires, or from human activities such as urban development, agriculture, and infrastructure construction.

Change detection in remote sensing is typically achieved through the comparison of pre-event and post-event satellite images. Spectral analysis of the images can be used to determine the areas where major changes have occurred. This process requires multiple approaches, such as pixel-based comparison, object-based analysis, and machine learning techniques that identify, classify, and measure these changes. High-resolution imagery and advanced processing methods have greatly contributed to the accuracy and robustness of remote sensing change detection, which is a necessary capability in both research and practice.

As satellite imagery has become more accessible, especially with the advent of Landsat, Sentinel, and commercial satellites, remote sensing change detection has become both more accessible and less expensive. Nonetheless, there remain difficulties in addressing image noise, data heterogeneity, and complicated change patterns, which can only be resolved with the help of sophisticated computational models. Such difficul-

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ties have stimulated the development of advanced algorithms and machine learning models, specifically convolutional neural networks (CNNs) and deep learning methods that can process high volumes of data and identify subtle shifts that might otherwise be overlooked by conventional approaches.

The primary contributions of this article are as follows:

Model Development: The article presents a novel deep learning-based system for identifying changes in satellite images, where a convolutional neural network (CNN) is used to learn the spatial features and patterns of the images. The model is trained using a large set of pre-event and post-event satellite imagery, enabling it to generalize across a diversity of landscapes and types of changes.

Enhanced Performance Measures: The model is evaluated using a comprehensive set of performance measures consisting of total loss, accuracy, precision, recall, F1-score, IoU (Intersection over Union), Dice loss, and BCE (Binary Cross-Entropy) loss. These indicators provide a thorough analysis of model performance and focus on aspects such as false positives, false negatives, and the general reliability of predictions.

Addressing Challenges in Change Detection: The article discusses common challenges in remote sensing change detection, including coping with image resolution variations, spectral variations, and noise in satellite images. Through the implementation of advanced image processing techniques and machine learning algorithms, the model demonstrates a higher capacity for dealing with these issues and achieving more accurate change detection results.

II. Literature Review

The literature review of advancements in change detection is categorized as follows:

A. Conventional Change Detection Approaches

Various methods of change detection in remote sensing have evolved over the past few decades, with many techniques being developed to address the problem of identifying changes in satellite or aerial images. Earlier techniques were mostly based on basic pixel-by-pixel comparison and image differencing algorithms [3]. Image differencing is one of the simplest change detection methods whereby differences between pre-event and post-event images are computed on a pixel-by-pixel basis [4]. The result is a difference image illustrating where major changes have occurred. This technique is, however, susceptible to noise and may not capture subtle changes, particularly where the spectral signature of the change resembles that of the background (e.g., seasonal changes or atmospheric effects).

Another classical method is Change Vector Analysis (CVA), which consists of computing the difference in a vector of corresponding pixels in multi-temporal images. CVA enables both the magnitude and direction of change to be identified, making it more effective than simple differencing techniques in capturing complex changes. Nonetheless, CVA can be computationally intensive and sensitive to image quality; therefore, preprocessing should be carefully performed to reduce inaccuracies caused by atmospheric effects or sensor artifacts.

Another common method is Post-Classification Comparison, in which each image is assigned to thematic categories (e.g., land cover types) and the classifications over time are compared to determine areas of change. Although this is an effective technique for identifying general land cover variations, it is highly sensitive to classification errors and the selection of the classifier. The technique also suffers from the salt-and-pepper effect, where the boundaries between classification codes are not necessarily smooth, resulting in spurious outcomes. Nevertheless, these limitations do not prevent the use of traditional methods, as they remain simple and easy to implement.

B. Modern Approaches and Machine Learning in Change Detection

As high-resolution imagery has become available and the volume of remote sensing data has become increasingly complex, current techniques of change detection have turned to more advanced approaches, frequently involving machine learning (ML) and deep learning (DL) methods. These methods have proven to be highly promising in addressing the shortcomings of conventional methods, particularly in processing very large datasets and identifying subtle and complex changes that were difficult to detect previously.

Among these approaches, the use of Convolutional Neural Networks (CNNs) is one of the most prominent, as the method can be applied to image analysis tasks and is especially efficient in learning the spatial features of input data independently [5], [6]. Pre-event and post-event images have been classified as change and no-change using CNNs. Such networks are capable of learning hierarchical characteristics and hence become resistant to image noise, different light sources, and complicated spatial structures. Deep learning frameworks for object detection and classification have demonstrated significant success in various computer vision applications [7]–[9], providing robust feature extraction capabilities that are transferable to change detection tasks. More recently, Generative Adversarial Networks (GANs) have been considered for change detection [10]. GANs are capable of creating high-quality synthetic images that recreate post-event imagery based on pre-event information, making it possible to compare the generated image with a real one to identify changes. This approach has shown promise in detecting subtle changes, particularly in regions where traditional methods fail [11].

Other machine learning algorithms that have been implemented for change detection tasks include CNNs and GANs, as well as Random Forests (RF) [12], Support Vector Machines (SVMs) [13], and k-Nearest Neighbor (k-NN). Advanced object detection frameworks [14], [15] and hybrid approaches [16] have contributed to improved accuracy in identifying and localizing changes in remote sensing imagery. Such techniques are typically applied together with image segmentation techniques, where images are broken into smaller, more manageable parts, and change detection is performed at the object or segment level. Such strategies are useful in minimizing the noise that may be experienced in pixel-based techniques

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and may provide more meaningful outcomes in complicated landscapes.

There has also been application of multispectral and hyperspectral data with machine learning algorithms to improve change detection. Multispectral information that records a wide spectrum of wavelengths provides more details on properties of the land surface compared to conventional RGB imagery. Recent advances in image processing techniques for underwater object detection [17] have demonstrated the effectiveness of deep learning approaches in challenging imaging conditions, with principles applicable to remote sensing scenarios. Hyperspectral sensors, which record hundreds of narrow spectral bands, provide even more detailed information, making it easier to differentiate between different types of changes, including vegetation stress or urban development. Integrating these data types with advanced machine learning methods has led to enhancement of the sensitivity and specificity of change detection models.

In addition, deep learning models for change detection have been developed to manage volumes of data effectively. These frameworks can acquire more complicated spatiotemporal patterns than conventional procedures. Methods such as U-Net for semantic segmentation and Fully Convolutional Networks (FCNs) have been used to identify changes at a pixel level and provide enhancements in accuracy and robustness. The application of machine learning extends beyond remote sensing to various domains including medical imaging [18], demonstrating the versatility and effectiveness of these computational approaches across different fields.

To conclude, change detection in remote sensing has experienced significant development, with machine learning and deep learning methods at the forefront. These contemporary techniques are not only more accurate and robust when it comes to identifying changes, but they also have the flexibility to work with the large and highly complex amounts of data that are common in remote sensing. With the ongoing development of technology, it is anticipated that more efficiency, interpretability, and scalability will be achieved in models, which will contribute to the further improvement of the potential uses of change detection in disaster response, land-use monitoring, and environmental management.

III. Proposed Model

The proposed model is a hybrid CNN-Transformer architecture shown in Fig. 1, designed for binary change detection from satellite imagery, with the following features:

A. Model Architecture

The architecture consists of the following key components:

U-Net Backbone: The encoder captures spatial information at different levels, while the decoder refines the spatial details.

CBAM Attention Module: The Convolutional Block Attention Module (CBAM) is integrated to enhance the model's focus on relevant regions of the image by applying attention mechanisms both in the channel and spatial dimensions. This helps in emphasizing important features and suppressing irrelevant ones.

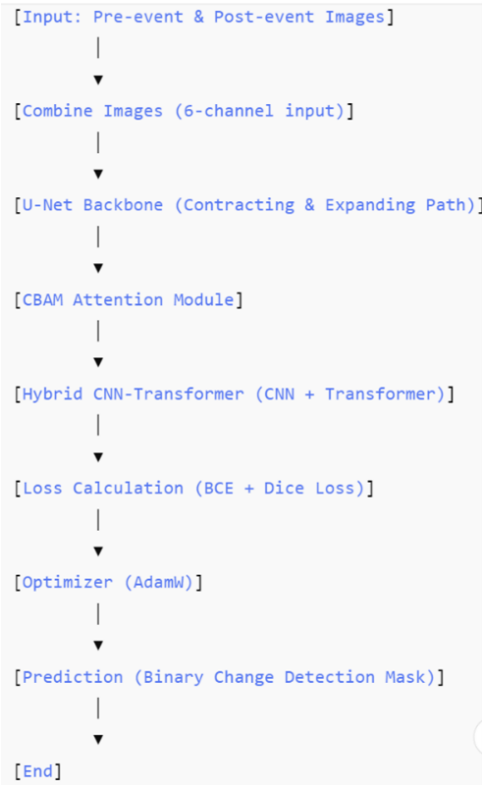


Fig. 1. Proposed change detection model architecture.

Input: The input to the model consists of 6-channel concatenated images: 3 channels from the pre-event image and 3 from the post-event image.

Output: The output of the network is a binary mask indicating areas of change (1) and no change (0).

B. Training Process

The model is trained through the following steps:

Loss Function: The model uses a combined loss function that incorporates Binary Cross-Entropy (BCE) and Dice Loss:

$$\text{Loss} = \lambda_{\text{BCE}} \cdot \text{BCE}(y_{\text{pred}}, y_{\text{true}}) + \lambda_{\text{Dice}} \cdot \text{Dice}(y_{\text{pred}}, y_{\text{true}}) \quad (1)$$

where y_{pred} is the predicted binary change map, y_{true} is the ground truth binary change map. BCE measures the pixel-wise difference between predicted and actual values. Dice Loss measures the overlap between predicted and actual change regions, promoting better segmentation.

Optimizer: The optimizer used is AdamW with learning rate scheduling to dynamically adjust the learning rate during training. The AdamW optimizer is preferred as it improves training stability with weight decay regularization. The update rule for AdamW is:

$$\theta_{t+1} = \theta_t - \eta \frac{m_t}{\sqrt{v_t} + \epsilon} - \lambda \cdot \theta_t \quad (2)$$

where m_t is the first moment estimate (mean), v_t is the second moment estimate (variance), η is the learning rate, λ is the

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weight decay factor, and ϵ is a small constant to prevent division by zero.

C. Model Execution and Evaluation

Prediction: The trained model generates a change detection mask indicating areas of change.

Metrics: The model's performance is evaluated using accuracy, precision, recall, F1-score, and IoU (Intersection over Union) metrics, which are standard for evaluating segmentation tasks.

D. Key Equations

Binary Cross-Entropy (BCE) Loss:

$$\text{BCE}(y_{\text{pred}}, y_{\text{true}}) = -\frac{1}{N}\sum_{i=1}^{N}[y_{\text{true}}\log(y_{\text{pred}}) + (1-y_{\text{true}})\log(1-y_{\text{pred}})] \tag{3}$$

where N is the number of pixels in the image, and y_{true} and y_{pred} are the ground truth and predicted labels, respectively.

Dice Loss:

$$\text{Dice}(y_{\text{pred}}, y_{\text{true}}) = \frac{2\sum_{i=1}^{N} y_{\text{pred}}^{i} y_{\text{true}}^{i}}{\sum_{i=1}^{N} y_{\text{pred}}^{i} + \sum_{i=1}^{N} y_{\text{true}}^{i}} \tag{4}$$

where N is the number of pixels, and y_{pred} and y_{true} are the predicted and ground truth values, respectively.

This model effectively combines the strengths of CNNs for spatial feature extraction and Transformer models (via attention mechanisms) for focusing on relevant areas of the image, which is ideal for change detection tasks that require distinguishing subtle changes in complex environments.

IV. Results

The results of this section are divided into different subsections as follows:

A. Pre-Event and Post-Event Image Comparison and Predictions

The pre-event and post-event satellite images, as well as the subsequent predictions, provide an elaborate examination of how the model has been able to identify variations across two different time scales (Fig. 2). The ground truth images represent the actual differences between the two time frames, where certain parts of the images show changes such as natural calamities or human interventions. These ground truth areas are then compared with the model's predictions, which attempts to identify similar areas where changes have occurred. The predictions are presented in the form of binary masks, where white areas indicate where change is predicted to occur. Probability maps are also presented in conjunction with these to visualize the certainty of the model's predictions. The probability maps use a color gradient to reflect certainty in the model, with higher values of certainty (values closer to 1) indicating high confidence that a change has occurred, and low values indicating uncertainty. These findings prove to be a graphical representation of the model's behavior in various locations, how the model detects differences, and the degree of certainty in its forecasts. The predictions are more accurate in some locations and less accurate in others, suggesting that the model may struggle with more subtle changes or challenging conditions, although the probability maps can be used to understand where the model is more reliable and where it may need additional training or modifications.

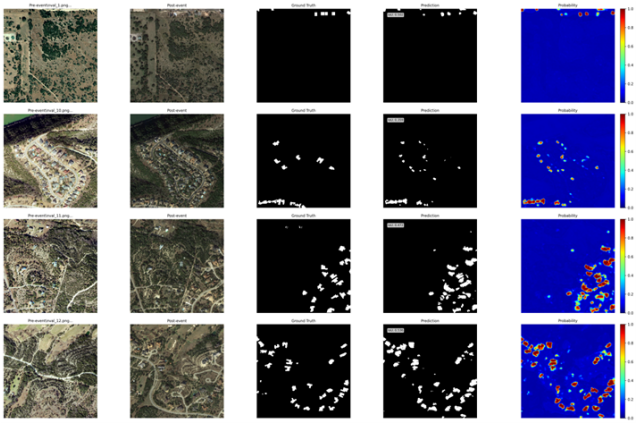


Fig. 2. Change detection using the proposed model.

B. Training and Validation Curves

The training and validation curves, which are provided for a variety of performance metrics in Fig. 3 (total loss, IoU, F1-score, accuracy, precision, recall, BCE loss, Dice loss), provide in-depth insights into the learning process of the model. The total loss and BCE (Binary Cross-Entropy) loss are used to indicate the model's capacity to differentiate between change and no-change; lower values indicate higher performance of the model. The IoU score is used to evaluate the correspondence between the predicted change areas and the ground truth, where an IoU of 0.614 was the best score, indicating that the model has a moderate correspondence between the predicted and actual changes. In detection tasks, precision and recall are especially significant because precision measures the relative number of correctly predicted changes among all predicted changes, and recall measures the number of changes that the model can accurately detect. The precision and recall curves show varying values, as the model is more likely to overestimate the occurrence of changes at approximately epoch 25, which leads to false positives. Recall indicates that the model is lacking in detecting some of the actual changes, as depicted in the F1-score, a balanced measure of precision and recall, with the highest score being 0.757. The Dice loss curve and BCE loss curve show how the model segments change regions, with Dice loss decreasing over time, which is expected for an effective segmentation task. These curves indicate that although the model performs well in most aspects, there is still room for improvement, particularly in recall and precision, which might result in better detection of changes.

C. Validation Set Confusion Matrix

The validation set confusion matrix further subdivides the model's performance by displaying the distribution of model

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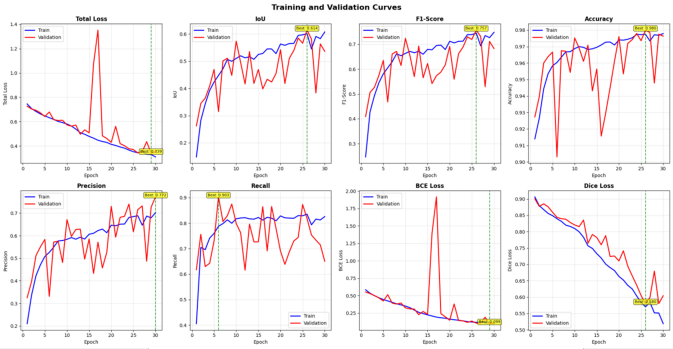


Fig 3. Proposed model evaluation metrics performance analysis

Fig. 3. Training and validation curves for various performance metrics.

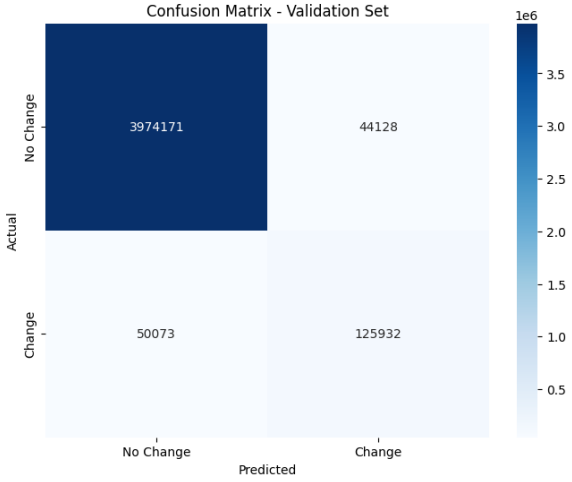


Fig. 4. Proposed model confusion matrix.

predictions in terms of true positives, false positives, true negatives, and false negatives (shown in Fig. 4). According to the matrix, the model performs best at predicting no changes (true negatives) with the highest number of 3,974,171 cases where the model correctly predicted areas that did not experience any changes. Nevertheless, there are also 44,128 false positives according to the matrix, where the model wrongly forecasted change in regions that did not actually change. Moreover, the number of false negatives is 50,073, where the model failed to detect genuine changes. Such false negatives imply that there are missed changes in the imagery, which is especially important in change detection tasks where it is essential to capture all instances of change. The 125,932 true positives show that the model has identified most of the areas where changes occurred, but the false positives and false negatives demonstrate the challenges the model faces in capturing all relevant changes while avoiding incorrect predictions. This distribution in the confusion matrix highlights the necessity to further optimize the model, especially with regard to precision and recall, to enhance the detection accuracy in case of changes.

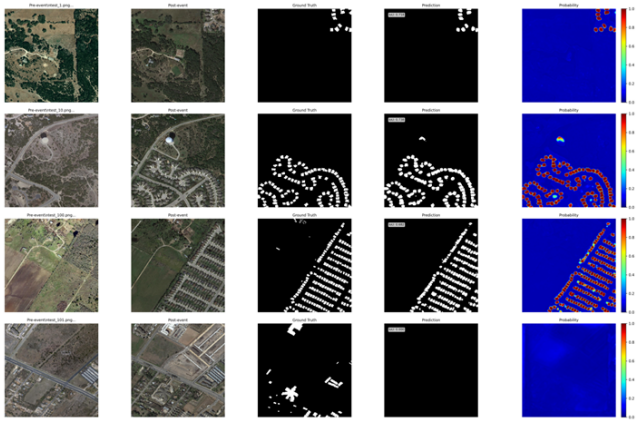


Fig. 5. Proposed model results for validation after testing.

D. Test Evaluation Results

The results of the test evaluation shown in Fig. 5 provide a detailed overview of the general performance of the model on the test set. The loss value of 0.3473 represents the extent to which the model generalizes to unknown data, with lower values depicting better performance. The BCE loss of 0.1094 indicates that the model performs well in distinguishing between change and no-change categories, as BCE loss is a major important parameter in binary classification tasks. The Dice loss of 0.5853 quantifies the model's capability to segment change areas, and a smaller value is a good indicator of effective segmentation performance. The accuracy of 97.32% is very high, which means that the model predicts most of the test images correctly. Precision (75.29%) and recall (70.57%) provide a more detailed picture: the model is quite effective at making correct predictions when it predicts change, but it misses some actual changes (lower recall) or predicts change when there is none (indicated by lower precision relative to accuracy). The F1-score of 72.86% offers a balance between precision and recall and is a better representation of the overall performance of the model. The IoU value of 57.30% indicates that the model can identify a moderate degree of overlap between the areas of predicted and actual changes, though there is room for improvement, particularly in terms of precisely defining change boundaries. Generally, the test evaluation results indicate that even though the model is currently performing well with high rates of accuracy, additional refinement is required to increase the model's capacity for detecting and predicting changes with high accuracy, particularly in problematic areas or regions with subtle changes.

V. Conclusion

The outcomes of this detection model demonstrate good performance in general, particularly in accuracy, precision, and Dice loss. Nevertheless, the confusion matrix and evaluation metrics analysis reveals areas for improvement, specifically in recall and precision. False positives and false negatives are significantly involved in change detection processes, as

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they may result in missed detections and false predictions. The training curves also indicate that there is a possibility of the model overfitting slightly, as the validation metrics fluctuate more than the training metrics. To enhance the performance of the model, additional hyperparameter tuning, model architecture modifications, or even more training data can be introduced to alleviate some of the problems identified with false positives and false negatives. The findings provide a foundation for the potential of the model and indicate where future initiatives can be concentrated to increase the accuracy and dependability of change detection in satellite images.

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